TP 1

Introduction to Network Security

1. Assume that you are the owner of ATM machine solution and you are requested to present your solution:
    a. What is the Attack Surface of ATM Machine?
    b. What is the Attack Services should be applied for ATM Machine?
    c. Draw your Cash Withdraw Security Model to represent its process
2. Assume that your system work as this function below:

```
import tkinter as tk
from tkinter import messagebox
import sys
# Simulated user database (username: password)
users = {
  'user1': 'abc',
  'user2': 'abc'
}
account = {
  '1234': 10000
}
def pop():
  money = int(entry_money.get())
  pin = entry_pin.get()

  if pin in account and account[pin] >= money:
    popup.withdraw()
    messagebox.showinfo("Withdraw Successful", "Thank you for using our service")
    sys.exit()
  else:
    messagebox.showerror("Withdraw Error", "Please try again.")

def login():
  username = entry_username.get()
  password = entry_password.get()

  if username in users and users[username] == password:
    root.withdraw()  # Hide the main window
    show_login_popup()
  else:
    messagebox.showerror("Login Failed", "Invalid credentials. Please try again.")

def show_login_popup():
  global popup
  popup = tk.Toplevel(root)
  popup.title("Cash Withdraw Page")
  global entry_money
```

```python
    global entry_pin
    label_money = tk.Label(popup, text="Input Your Money:")
    label_pin = tk.Label(popup, text="Input Your Pin Code:")
    entry_money = tk.Entry(popup)
    entry_pin = tk.Entry(popup, show="*")
    button_pop = tk.Button(popup, text="Okay", command=pop)
    label_money.grid(row=0, column=0, padx=50, pady=50)
    entry_money.grid(row=0, column=1, padx=50, pady=50)
    label_pin.grid(row=1, column=0, padx=50, pady=50)
    entry_pin.grid(row=1, column=1, padx=50, pady=50)
    button_pop.grid(row=2, columnspan=2, padx=50, pady=50)
    popup.mainloop()

root = tk.Tk()
root.title("Welcome To Our ATM System")
label_username = tk.Label(root, text="Username:")
label_password = tk.Label(root, text="Password:")
entry_username = tk.Entry(root)
entry_password = tk.Entry(root, show="*")
button_login = tk.Button(root, text="Login", command=login)
label_username.grid(row=0, column=0, padx=50, pady=50)
entry_username.grid(row=0, column=1, padx=50, pady=50)
label_password.grid(row=1, column=0, padx=50, pady=50)
entry_password.grid(row=1, column=1, padx=50, pady=50)
button_login.grid(row=2, columnspan=2, padx=50, pady=50)
root.mainloop()
```

- What is the solution to prevent multiple attempts?